ITM SKILLS UNIVERSITY

School of Future Tech

Case Study Number 76 – Report

on

CraftCircle — DIY Workshop Portal

by

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1. Introduction to the Case Study

Creative workshops such as pottery, painting, candle making, and crochet have become increasingly popular among students and hobbyists. These workshops allow individuals to explore artistic creativity while learning practical skills. However, many small creative studios still rely on manual methods for managing workshop registrations and schedules.

CraftCircle is a web-based platform designed to simplify the process of discovering and booking creative workshops. The platform allows users to explore different workshop categories, view schedules, and reserve seats online.

This case study focuses on the design and development of the CraftCircle platform using modern web technologies. The project demonstrates how a well-structured website can improve user experience while simplifying the workshop registration process.

Key Features at a Glance

- Workshop listings — browse workshops by category including pottery, painting, candle making, and crochet.
- Booking system — fill out a booking form and select workshop sessions with ease.
- Interactive seat selection — a visual seat map allows users to choose available seats.
- Materials information page — guidelines and material lists provided for each workshop.
- Responsive interface — layout adapts automatically for desktop and mobile screens.
- Form validation — JavaScript validates all user inputs before confirming a booking.
- LocalStorage persistence — all booking data stored in the browser, no backend required.
- Calendar view — upcoming workshop sessions displayed in a structured calendar layout.

2. Problem Statement / Case Background (Abstract)

Background

Many creativestudios manage workshop registrations manually using spreadsheets, phone calls, or social media messages. These methods often lead to problems such as scheduling conflicts, lack of proper records, and difficulty tracking seat availability.

A digital platform that centralises workshop information and booking functionality can solve these issues by providing a structured system for managing workshop sessions. The key pain points in the current landscape include:

- Manual registration — studios rely on phone calls and spreadsheets, causing errors and double-bookings.
- No seat tracking — no real-time visibility into which seats are available for a given session.
- Scheduling conflicts — without a centralised system, clashes between sessions are common.
- Poor discoverability — students and hobbyists have no single place to browse local workshops.
- No materials guidance — participants often arrive unprepared due to lack of pre-session information.

Abstract

This case study presents the development of CraftCircle, a DIY workshop booking platform that enables users to browse creative workshops and reserve seats online. The platform includes multiple sections such as a home page, workshop listings, booking interface, calendar view, and materials information page.

The system is built using HTML, CSS, and JavaScript, focusing on responsive design and interactive features. Through this platform, users can easily explore workshops and book sessions while the system manages seat availability and booking details. All data is persisted using the browser's localStorage, making the application fully deployable as a static site with no backend infrastructure required.

3. Case Study Design

CraftCircle is designed as a modular web application consisting of several interconnected pages, each with clearly defined responsibilities to ensure maintainability and a clean separation of concerns.

System Architecture

Layer	Component / Description
Presentation Layer	HTML pages displaying the interface CSS files
Styling Layer	controlling layout and design JavaScript handling
Interaction Layer	dynamic behaviour Browser localStorage used to
Data Layer Media	simulate booking storage Images used for workshop
Layer	display cards

Application Flow

1. User visits the Home Page and views featured workshops.
2. User navigates to the Workshops Page to browse available sessions.
3. User selects a workshop and proceeds to the Booking Page.
4. User enters personal details and selects a workshop session and seat.
5. The system validates the form and displays a booking confirmation message.

This workflow provides a simple and user-friendly booking experience that requires no technical knowledge from the end user.

Data Model

All booking data is stored in the browser's localStorage. The key storage entries used by the application are:

localStorage Key	Contents
craftcircle_bookings	JSON array of all booking objects — workshop ID, user details, selected seat, session, and timestamp.
craftcircle_seats	JSON object keyed by workshop ID, tracking which seat numbers are occupied per session.

4. Methods & Technology Applied

4.1 Technology Stack

Category	Technology
Frontend	HTML5
Structure	CSS3
Styling	JavaScript (ES6)
Programming Language	CSS Grid, Flexbox
Layout Techniques Fonts	Google Fonts
Deployment	craftcirclestudio.vercel.app
Browser API	localStorage

4.2 Responsive Design

TheplatformusesCSS Gridand Flexbox to ensure that the layout adapts to different screen sizes. Media queries adjust column counts, font sizes, and spacing for tablet and mobile viewports, ensuring a consistent experience across devices.

4.3 Interactive Booking System

JavaScriptdynamically updates seat availability and booking details based on user interaction. When a user selects a session, the seat map re-renders to reflect only the available seats for that specific session, reading from and writing to localStorage in real time.

4.4 Image-Based Workshop Cards

Each workshopi s displayed using a visual card containing an image, title, description, duration, and price. Cards are rendered dynamically from a JavaScript data array, making it straightforward to add or remove workshops without modifying the HTML markup.

4.5 Form Validation

Userinput fields arevalidated using JavaScript before confirming a booking. Validation checks include required field presence, valid email format, and ensuring a seat has been selected before the form is submitted. Error messages are displayed inline next to each invalid field.

4.6 Smart Chart Selection Algorithm

Data / Content	Display Method	Rationale
Seat availability per session	Interactive seat grid	Visual, colour-coded occupied vs. available seats
Workshop category breakdown	Card-based listings	Image cards group workshops by type for easy browsing

5. Implementation Details

5.1 Project File Structure

File / Folder	Purpose
index.html	Home page of the website
workshops.html	Displays all available workshops
booking.html	Workshop booking interface
calendar.html	Shows upcoming workshop sessions
materials.html	Displays workshop materials and preparation guidelines
shared.css	Global styling used across all pages
index.css	Styling specific to the home page
workshops.css	Styling for workshop listing page
booking.css	Styling for the booking interface
materials.css	Styling for the materials page
shared.js	Navigation and shared interactive functionality
Images/	Folder containing workshop images
documentation/	Folder containing project documentation Project
README.md	description and setup instructions

5.2 Key Implementation Decisions

Multi-PageApplicationwithouta Framework: CraftCircle uses separate HTML files for each major section rather than a single-page app.

This simplifies the routing model and keeps each page's code self-contained, which is well-suited for a static deployment.

Data-Driven Workshop Cards: Workshop listings are generated from a central JavaScript data array rather than being hard-coded in HTML. This separation of data and presentation makes it easy to update workshop details without touching the markup.

Seat Map Implementation: The seat selection grid is rendered as a table of buttons. Each button reads its state from localStorage on page load and toggles between available and selected states on click, updating the stored seat data immediately.

Shared Navigation: The shared.js file injects the navigation bar into every page programmatically, ensuring a consistent header without duplicating HTML across files.

5.3 Key Features Implemented

Feature	Description
Workshop Listing	Users can browse different workshop categories such as pottery, painting, candle making, and crochet.
Booking System	Users fill out a booking form and select workshop sessions through a clean interface.
Seat Selection	An interactive seat map allows users to select available seats for a session.
Materials Information	The materials page lists all supplies provided and preparation guidelines per workshop.
Responsive Interface	The website adjusts its layout automatically for desktop and mobile screens.

6. Results and Conclusion

Findings

TheCraftCircle platform successfully demonstrates a digital system for managing creative workshop bookings. The following outcomes were observed across development and testing:

- Structured navigation across multiple pages with a consistent shared header.
- Interactive booking and seat selection system functioning correctly with localStorage persistence.
- Image-based workshop display cards rendering responsively across screen sizes.
- Form validation correctly preventing incomplete or invalid submissions.
- Materials page providing clear per-workshop preparation guidelines.
- Performance: as a fully static, no-backend application, page load times remain under one second.

Limitations

- localStorage is browser-scoped: booking data is isolated to each browser and device instance. A production deployment would require a backend database.
- No user authentication: any user can access all bookings. A login system would be required for real-world use.
- No payment integration: the booking flow does not process payments, limiting deployment to pilot or internal use cases.
- No real-time collaboration: multiple staff members cannot manage the same workshop schedule simultaneously.

Conclusion

CraftCircle demonstrates that a fully-featured workshop discovery and booking platform can be built and deployed without any backend infrastructure using modern browser APIs. The project combines responsive design, interactive JavaScript, and clean UX into a cohesive product that directly addresses the manual booking pain points faced by small creative studios.

The architecture — a multi-page HTML application with localStorage persistence and a data-driven component model — proves that sophisticated web applications do not require complex frameworks or server infrastructure to deliver a high-quality user experience. The platform is immediately usable by any user with a browser.

Future improvements may include Firebase or Supabase integration for persistent cross-device storage, user authentication to prevent duplicate bookings, an integrated payment gateway, and a mobile-first redesign to better serve smartphone users.

7. References

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6. CSS Grid Layout Guide – https://css-tricks.com/snippets/css/complete-guide-grid/

7. Flexbox Guide – https://css-tricks.com/snippets/css/a-guide-to-flexbox/

8. Online tutorials and web development articles related to responsive design and vanilla JavaScript.

Declaration

I hereby declare that this case study report on CraftCircle — DIY Workshop Portal is my original work. All references and resources used in this report have been properly cited. The application, its code, and the analysis presented in this report are my own work unless otherwise stated.

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